PolicyLens: Fault-Tolerant Regulatory Ingestion, Lifecycle-Aware Indexing & Grounded RAG

Author: Vinit Surve | **Domain:** Indian Financial Regulation (RBI, SEBI, IRDAI, DFS) | **Date:** November 2026

Executive Abstract

Navigating dense, continually updated regulatory frameworks promulgated by financial oversight bodies—such as the Reserve Bank of India (RBI), Securities and Exchange Board of India (SEBI), and Insurance Regulatory and Development Authority of India (IRDAI)—poses an acute operational challenge for automated compliance systems. Conventional Retrieval-Augmented Generation (RAG) architectures exhibit catastrophic failure modes in compliance domains: vector store contamination from repealed or superseded circulars, asynchronous worker write races during batch indexing, collapsed two-column reading flows, flattened table schemas, and regulatory hallucinations. PolicyLens is an enterprise-grade regulatory document ingestion, indexing, and question-answering architecture providing enforced lifecycle consistency and grounded retrieval with provenance citations. Evaluated across an audited corpus of **40 official regulatory documents (1,123 pages; 1,604 active chunks in production collection policylens_v1)**, PolicyLens achieves a **96.0% Recall@3** and **0.9100 MRR** in controlled chunking comparisons against fixed-window baselines, and **84.0% Recall@3** and **0.7657 MRR** (**96.0% Recall@10**) in full-corpus unrestricted global retrieval on the 1,662-chunk revised validation index via entity-aware hybrid lexical reranking with zero regressions, maintaining **10.3 ms mean vector query latency** on local CPU hardware with **zero external API embedding costs**.

1. Problem Statement & Legal Complexity

Indian financial regulations are versioned statutory instruments with complex structures. Naive RAG pipelines break down across five distinct failure modes: (1) *Stale Vector Contamination* where repealed circulars remain searchable; (2) *Worker Write Races* where late batches from failed workers corrupt production indices; (3) *Two-Column Layout Disruption* where horizontal scanning concatenates disjoint columns; (4) *Table Schema Collapse* where merged cells and financial fee brackets are scrambled; and (5) *Regulatory Hallucination* where language models invent circular numbers or interest rates.

2. High-Level System Architecture

Pipeline Stage	Core Component	Engineering Mechanism & Invariant
1. Parsing	PyMuPDF + Tesseract	Two-column gutter detection; selective OCR on raster scans only; table extraction via pdfplumber.
2. Chunking	RegulatoryChunker	~350 body tokens with breadcrumbs [Doc > Section]; Markdown table schema preservation; adjacent chunk expansion.
3. Embedding	SentenceTransformers CPU	all-MiniLM-L6-v2 (384-dim, normalized). Runs entirely local on CPU at zero external API cost.
4. Lifecycle	SQLite State Machine	Fenced worker leases, PRAGMA foreign_keys=ON, atomic Compare-And-Swap (CAS) candidate promotion.
5. Search & Retrieval	ChromaDB + Entity Rerank	Guarded 4-field pre-validation; isolated active fingerprint filter; entity-aware hybrid Config 4 rerank.
6. Grounded QA	RegulatoryQAEngine	Strict compliance prompt; exact provenance citations; zero-LLM honest fallback if score < 0.25.

3. Regulatory Corpus Audit & Structural Complexity

A rigorous physical audit was conducted across all 40 official PDF documents catalogued in data/manifest.csv. The corpus totals **1,123 pages (30.75 MB)** across four major Indian regulatory authorities. 100% of URLs originate from verified government domains with zero corrupt downloads or missing metadata fields.

Regulatory Authority	Abbr.	Docs	Pages	Share	Core Subject Coverage
Reserve Bank of India	RBI	22	731	65.1%	Pension Disbursement, Internal Ombudsman, IRAC, Basel III
Dept. of Financial Services	DFS	9	202	18.0%	Gazette Notifications, NaBFID Rules, PSB Board Appointments
Securities and Exchange Board	SEBI	8	174	15.5%	NRI KYC Simplification, Currency Derivative Limits, BCP
Insurance Regulatory Authority	IRDAI	1	16	1.4%	Registration, Capital Structure, Transfer of Insurer Shares
Corpus Total	**ALL**	**40**	**1,123**	**100%**	**Comprehensive Indian Financial Oversight Umbrella**

4. Parsing Quality & Scanned Page OCR Accuracy

To evaluate layout parsing fidelity, 10 representative pages across native text, multi-column articles, complex tables, and raster Gazette scans were manually inspected. Native digital text extracted with 100% precision. On pure scanned raster pages lacking digital text streams (such as dfs_sidbi_dmd_2026.pdf), PolicyLens triggers Tesseract OCR fallback dynamically.

Scanned OCR Case Study: dfs_sidbi_dmd_2026.pdf (Page 1 Gazette Scan)

- **Native Text Detected:** 0 bytes (pure raster image at 200 DPI).
- **OCR Engine & Latency:** Tesseract 5.5.0 executed in **0.91 seconds** across 8 blocks.
- **Measured Word Accuracy on Typed Legal Body: 95.8% (91 / 95 words)**.
- **Word Substitutions Observed:** Minor character slips ('Part II' for 'Part II', '3"' for '3rd').
- **Handwritten Signature Handling:** Officer signature seal correctly isolated into a low-confidence terminal block, preventing noise leakage into body search chunks.

5. The 10-Step Safe Lifecycle & CAS Promotion Protocol

A core contribution of PolicyLens is eliminating stale vector retention and concurrent worker write races by modeling document ingestion as an ACID-compliant state machine governed by SQLite and enforced via database triggers.

Step	Ingestion Phase	State & Invariant Enforced
1–2	Parsing & Layout Detection	PyMuPDF extracts blocks, pdfplumber renders Markdown tables, gutter analysis detects columns.
3–4	Chunking & Fingerprinting	Structure-aware chunks (~350 body tokens + breadcrumbs). Deterministic SHA-256 fingerprint computed.
5	Idempotency Check	If index_fingerprint matches active_fingerprint in document_registry, returns SKIPPED in 2.1 ms.
6	Staging Registration	SQLite persists expected_active_fingerprint, fencing_token=1, host_id, pid. State: STAGING.
7	Local CPU Embedding	SentenceTransformers encodes batch on local CPU into 384-dimensional normalized float vectors.
8	Guarded ChromaDB Ingestion	Validates doc_id, collection, fingerprint, version_id on all chunks before calling Chroma.add().
9	Staging Verification	Worker confirms chunk and token count match SQLite registration. State transitions: VERIFIED.
10	Atomic CAS Promotion	CAS check: expected_active_fingerprint == current_active. Promotes candidate to ACTIVE; demotes prior active to SUPERSEDED. If stale, rejects with StalePromotionError.

6. State Transition Controls & Invariants

State transitions are constrained by SQLite database trigger trg_enforce_lifecycle_transition. Illegal transitions (such as direct revoked -> active or failed -> active) are aborted at the database engine level with RAISE(ABORT), preventing bypass by client application bugs.

Allowed Lifecycle Transitions:

- staging -> verified, staging -> failed, staging -> revoked
- verified -> active (strictly via atomic CAS), verified -> failed, verified -> revoked
- active -> superseded (automatic upon successful CAS promotion of newer version)
- revoked, failed, superseded are **terminal states** (no transitions to active permitted).

7. Systematic Retrieval Evaluation: Chunking & Global 40-Doc Configurations

To prevent conflation across candidate pool scales, PolicyLens evaluations are partitioned into two controlled studies over **25 realistic compliance questions** (14 standard legal text clauses, 6 table-lookup questions, and 5 paraphrased questions): (1) *Pilot Chunking Evaluation* comparing Structure-Aware against a Fixed-Window baseline (500 tokens, 50 overlap); and (2) *Full 40-Document Global Benchmark* across the 1,662-chunk disposable validation index comparing Dense, Config 4 Lexical Rerank, and Entity-Aware Hybrid retrieval without altering the 1,604-chunk production index policylens_v1.

Evaluation Framework & Configuration	Corpus Scope	Recall@3	Recall@5	Recall@10	MRR
Fixed-Window Baseline (500/50)	Pilot (6 docs, 45 chk)	92.0%	100.0%	100.0%	0.8333
Structure-Aware Chunking (Pilot)	Pilot (6 docs, 114 chk)	**96.0%**	100.0%	100.0%	**0.9100**
Dense Baseline (40-Doc Global)	Validation (1,662 chk)	64.0%	72.0%	88.0%	0.6285
Config 4 Lexical Reranking (Global)	Validation (1,662 chk)	72.0%	76.0%	92.0%	0.6614
Entity-Aware Hybrid + Config 4 (Global)	Validation (1,662 chk)	**84.0%**	**84.0%**	**96.0%**	**0.7657***
Entity-Aware Hybrid (Document Scoped)	Validation (Scoped)	**100.0%**	100.0%	100.0%	**0.9067**

**Note: 0.7657 reflects strict single-document gold matching (Q13 unranked). Under semantic multi-framework statutory matching (identical barred-complaint clause in rbi_io_re_2023), MRR is 0.8057 and Recall@3 is 88.0%.*

8. Disagreement, Disambiguation & Statutory Overlap Analysis

Targeted investigation reveals why entity disambiguation and structure-aware parsing are mandatory in multi-framework corpora:

QID & Focus	Question Text & Target Regulatory Clause	Dense	Final	Root Cause, Disambiguation & Strategic Invariant
Q06 Table Schema	Repealed circular dated 14.10.2009 for pension withdrawal facility (rbi_pension_2026, p.19).	Rank 4 (0.542)	**Rank 1** (0.658)	**Structure-Aware Win:** Preserves Markdown table schema with headers; fixed window severed header from row.
Q10 Sibling Circulars	Minimum threshold of banking outlets for commercial banks under Internal Ombudsman (rbi_io_comm_2026).	Unranked (>10)	**Rank 2** (0.701)	**Entity Disambiguation Win:** 5 sibling circulars (SFB, NBFC, PPI, CIC) share identical text; classifier prioritizes commercial banks.
Q23 Table Lookup	Agency bank listed under item 2 in Annex 1 listing (rbi_agency_comm_2026, p.16: Bandhan Bank).	Unranked (>10)	**Rank 3** (0.635)	**Config 4 Win:** Excludes domain stopwords ('bank', 'agency') and single digits ('2') from over-boosting narrative formats.
Q13 Statutory Overlap	Categories of complaints explicitly barred from being entertained by Internal Ombudsman.	Rank 1 (re_2023)	Rank 1 (re_2023)	**Statutory Overlap:** Gold labels comm_2026; retrieval returns parent re_2023 containing verbatim identical text. Strict gold preserved.

9. Efficiency, Robustness & Verified Production Statistics

Operational efficiency and correctness were measured across live ingestion, idempotency tests, and local CPU execution:

Performance Dimension	Measured Result	Benchmark Significance & Hardware Context
Full-Corpus Ingestion (Run 1)	**187.87 s (~3.1 min)**	End-to-end parse, OCR, embedding, and CAS promotion across all 40 PDFs / 1,123 pages (16.73 s / 100 pages).
Idempotent Re-Run (Run 2)	**0.09 s (90 ms)**	**2,087x speedup** across all 40 documents; exactly 0 new chunks added; zero duplicate vectors created.
Active Collections & Chunks	**1,604 Chunks**	policylens_v1 holds 1,604 active chunks (1,763 chunks total across collections) strictly isolated by fingerprint.
40-Doc Global Retrieval MRR	**0.7657 (Strict) / 0.8057 (Sem)**	**84.0% Recall@3**, **96.0% Recall@10** across full 40-PDF corpus with **0 regressions** across all 25 queries.
Warm Retrieval Latency	**10.32 ms (Mean)**	Sub-15ms vector query response (min: 9.8 ms, p95: 14.2 ms) over repeated runs on commodity CPU.
Storage Footprint	**28.25 MB Vector / 112 KB DB**	ChromaDB HNSW persistent vectors + SQLite ACID lifecycle registry covering all 40 regulatory circulars.
Local Embedding Cost	**$0.00 (Zero API Cost)**	Document embeddings are generated locally, without transmitting document content to an external embedding API.
Automated Test Suite	**101 / 101 Passing**	Executed locally via pytest across 6 test modules (101 collected, 101 passed, 5 warnings) using in-memory fixtures.

10. Implemented Security Controls vs. Proposed Enterprise Extensions

• **Implemented & Verified Controls:** Fingerprint-scoped vector isolation (where={'index_fingerprint': {'$in': active_fps}}), guarded 4-field metadata pre-validation before ChromaDB writes, SQLite ACID lifecycle state machine with foreign keys and database triggers, worker leases with monotonic fencing tokens, adjacent chunk expansion, local CPU embeddings (no external embedding API transmission), and honest fallback for out-of-scope queries.
• **Proposed Enterprise Scaling Roadmap:** Clustered vector database sharding (Milvus / Qdrant) partitioned by regulator and year for collections exceeding 100,000 documents; distributed lock management via Redis Redlock; multi-tenant RBAC access tiers (public_regulatory, internal_compliance, board_confidential) enforced at retrieval time.

11. Conclusion & Submission Readiness

PolicyLens bridges the critical chasm between research-grade RAG and production legal compliance. By formalizing document ingestion as an ACID-compliant lifecycle state machine, enforcing guarded vector writes, isolating retrieval to active versions, and utilizing structure-aware chunking with entity disambiguation, PolicyLens is designed to reduce unsupported regulatory answers, maintaining grounded citations and deterministic auditability.